

A Lightweight Causal Decoder Improves EEG Foundation Models for Real-Time Sleep Staging

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Abstract—State-of-the-art automatic sleep staging is dominated by two model families that are both incompatible with real-time use: bidirectional sequence models that read up to 35 epochs of *future* context, and large EEG foundation models that classify each 30-second epoch *independently*. We show that a single lightweight recipe—a causal transformer decoder (3.5–11M parameters) over per-epoch features—bridges both gaps. Trained on fixed spectral tokens, the decoder is competitive with billion-token foundation models on the standard HMC benchmark at $\sim 1/100$ th their size. Stacked on *frozen* features from the strongest foundation model (REVE), it produces the best published-protocol HMC result on every metric (balanced accuracy 75.1, κ 0.705, weighted-F1 77.0 over 8 seeds), surpassing REVE’s own full fine-tuning while training only the decoder—and, unlike every model above it, it emits decisions causally at one-epoch latency. Controlled experiments across three corpora (Sleep-EDF-78, HMC, Physio2018; published splits throughout) further show that (i) the cost of causality is ≈ 1 accuracy point, not the 3–7 points leaderboard gaps suggest; (ii) in-domain self-supervised pretraining of the decoder is null even with 994 training subjects; and (iii) pretraining pays only as a carrier of *out-of-domain* structure—cross-corpus transfer on simple features (+3.1) and foundation-model features are substitutes, not additive.

Index Terms—sleep staging, EEG, foundation models, causal transformers, real-time, self-supervised learning

I. INTRODUCTION

Automatic sleep staging assigns one of five stages (W, N1, N2, N3, REM) to each 30-second EEG epoch. The best published systems are *offline*: SeqSleepNet-lineage models [1], [2], [3], [4] classify an epoch from a bidirectional window of 10–35 neighbouring epochs, while U-Time-style architectures segment whole nights at once [5], [6]. A newer family of EEG *foundation models*—BIOT [7], LaBraM [8], EEGPT [9], CBraMod [10], NeuroLM [11], and most recently REVE [12], pretrained on 60,000 hours of EEG—approaches the same benchmarks from the opposite direction: enormous pretrained epoch encoders that classify each epoch *with no inter-epoch context at all*. Neither family can drive the growing set of applications that need staging *as the night unfolds*: closed-loop stimulation, on-device monitoring, and adaptive interventions.

This paper asks how much of the offline systems’ advantage actually requires their machinery, and answers with a deliberately small model: PhysioFM-S, a causal transformer decoder over per-epoch feature tokens, fine-tuned end-to-end. Our contributions:

- 1) **A strong real-time operating point.** On the fixed published HMC split, PhysioFM-S over 64-bin spectral tokens reaches balanced accuracy (BAC) 73.8 / κ 0.668 (8 seeds)—above all but one published EEG foundation model at $\sim 1/100$ th the parameters—while emitting decisions causally with $O(1)$ compute per epoch.
- 2) **Improving a foundation model without fine-tuning it.** Stacking the decoder on *frozen* REVE features yields the best published-protocol HMC result on every metric, surpassing REVE’s own LoRA fine-tuning (10 runs, model souping) while training only an 11M-parameter decoder. A context-free probe control attributes the entire gain to inter-epoch sequence modeling.
- 3) **The price of causality is ≈ 1 point.** Giving our model the offline systems’ bidirectional sequence stage—including a receptive-field-matched variant—moves results by ≤ 1 accuracy point across three corpora, and a fully bidirectional twin gains only +0.9 offline while losing 5.0 online. The remaining gap to offline SOTA is localized in the epoch-level feature extractor, not in missing future context.
- 4) **When pretraining pays.** Across 20+ controlled pre-training comparisons: in-domain predictive-coding pre-training of the decoder is null everywhere (even at 994 subjects); cross-corpus transfer on simple features is robustly positive (+3.1 $\pm$ 0.3); and on foundation-model features the transfer effect vanishes—the two mechanisms are substitutes, both supplying out-of-domain structure.

II. METHOD

A. Tokens

A recording is a sequence of 30-s epochs. Each epoch becomes one token: **(a) Spectral (tf64)**: per channel, 64 log-spaced Welch log-power bins (0.5–49 Hz, 4-s sub-windows), flattened over C channels; normalized by per-(channel, bin) statistics fixed on the training corpus (no per-recording normalization). **(b) Foundation (REVE)**: the epoch is preprocessed exactly as in REVE’s released pipeline and passed through the *frozen* REVE encoder [12]; per-channel mean over its 1-s patch embeddings gives a $C \times D$ token ($D=512$ Base, 1216 Large). **(c) Per-electrode**: variant of (a)/(b) with one sequence per

channel (tokens $1 \times D$); channel-count-agnostic, which enables full-weight transfer between corpora with different montages; per-epoch predictions average over channels.

B. Causal decoder

Tokens pass a linear embedding into a 6-layer causal transformer decoder (width 256, 8 heads, RoPE; 2.4–11M parameters depending on token width, built from TimesFM-2.5 decoder blocks [13]), followed by a linear classifier on each epoch’s hidden state. Attention is strictly causal: the prediction for epoch t uses epochs $\leq t$ only, so inference streams with a KV-cache at one token per decision. For analysis we also train (i) a bidirectional twin (identical stack, full attention) and (ii) a bidirectional *context head* (2-layer transformer, 1.1M parameters, optional band-limited attention) on top of the causal features, replicating the sequence stage of offline SOTA systems.

C. Training and pretraining

Fine-tuning is end-to-end with class-balanced cross-entropy (we also report plain cross-entropy where the comparison ladder uses it), 8–20 epochs, best epoch by validation κ where the published protocol selects on validation. Self-supervised *predictive-coding* (PC) pretraining trains the same decoder to regress the next 16 tokens (multi-step MSE) on unlabeled sequences; every pretrained model is paired with a random-initialization control under an identical schedule. Cross-corpus *transfer* pretrains on a donor corpus and fine-tunes on the target; per-electrode tokens allow moving *all* weights across montages. Pretraining corpora never contain target-corpus test subjects.

III. EXPERIMENTAL SETUP

Corpora and protocols (all published splits). *HMC* [14]: 151 clinical PSGs, 4 EEG @256 Hz; fixed split (100/25/26 recordings) and metrics (BAC/ κ /weighted-F1, pooled test epochs) of the NeuroLM protocol [11], reproduced bit-exactly (137,243 epochs). *Physio2018* [15]: 994 labeled records, 6 EEG @200 Hz; SleepPyCo’s released 5-fold subject-disjoint assignment [4], metrics pooled over all test epochs (accuracy/MF1/ κ). *Sleep-EDF-78* [16]: 153 recordings, 2 EEG @100 Hz; 5-fold subject-disjoint CV, standard ± 30 -min trim. REVE’s pretraining corpus excludes HMC recordings and contains neither Sleep-EDF nor Physio2018 [12], so no stacked result inherits leakage. Results average 3–8 seeds (seed-matched pretrain/fine-tune); mean $\pm$ sd reported.

IV. RESULTS

A. A small causal model on the foundation-model ladder

Table I: trained from scratch on one corpus, PhysioFM-S over spectral tokens is second of seven on BAC among published foundation models. Stacked on frozen REVE features, the decoder surpasses *every* published row: +1.1 to +1.8 BAC over REVE’s own fine-tuning, and under the ladder’s plain-CE convention it leads on all three metrics (κ 0.705 $\pm$ 0.006, wF1 77.0). The gain is attributable: a per-epoch probe on the

TABLE I
HMC, FIXED PUBLISHED SPLIT (TEST: 26 SUBJECTS / 23,871 EPOCHS).
FM = PRETRAINED EEG FOUNDATION MODEL (FULL FINE-TUNE). OUR
ROWS: 8 SEEDS. [†]PLAIN-CE ARM (LADDER LOSS CONVENTION);
[‡]CLASS-BALANCED ARM.

Model	BAC	κ	wF1
NeuroLM-B [11] (FM)	.674	.619	.713
BIOT [7] (FM)	.686	.630	.709
EEGPT [9] (FM)	.703	.658	.732
CBraMod [10] (FM)	.727	.669	.740
LaBraM-Base [8] (FM)	.729	.681	.755
REVE-Base full fine-tune [12] (FM)	.740	.698	.764
PhysioFM-S tf64, from scratch (3.5M)	.738	.668	.745
frozen REVE-B, per-epoch probe (control)	.724	.649	.727
frozen REVE-B + PhysioFM-S decoder [‡]	.752	.684	.756
frozen REVE-L + PhysioFM-S decoder [‡]	.758	.695	.765
frozen REVE-L + PhysioFM-S decoder [†]	.751	.705	.770

TABLE II
PHYSIO2018 (SLEEPYCO’S EXACT FOLDS; POOLED 892,200 TEST
EPOCHS) AND SLEEP-EDF-78 (5-FOLD). PUBLISHED OFFLINE MODELS
USE BIDIRECTIONAL MULTI-EPOCH CONTEXT.

Corpus	Model	acc	MF1	κ
P2018	U-Time [5] (offline)	78.8	77.4	.714
	SeqSleepNet [1] (offline)	79.4	77.6	.719
	SleepPyCo [4] (offline)	80.9	78.9	.737
	PhysioFM-S tf64 (causal)	76.1	75.2	.684
	frozen REVE-B + PhysioFM-S (causal)	78.3	77.3	.710
SEDF-78	XSleepNet2 [2] (offline)	84.0	77.9	.778
	PhysioFM-S tf64 (causal, 3 seeds)	78.7 $\pm$ 0.2	74.4	.713
	+ P2018 transfer (per-electrode)	78.4 $\pm$ 0.4	74.5	.712

same frozen features (no sequence context) scores 4–5 BAC points lower. REVE classifies each epoch independently; a 45-second-per-seed decoder training supplies what its 60,000-hour pretraining does not—and the composite remains causal, streaming at one-epoch latency, whereas every other row in the table is offline.

B. Large supervised benchmarks and the cost of causality

Table II: on Physio2018 the stack reaches 78.3/ κ 0.710—past U-Time, 2.6 points from SleepPyCo—and on Sleep-EDF 78.7/ κ 0.713, 5.3 from XSleepNet2. Every model above ours is bidirectional and offline. Is the gap the missing future context? No: (i) adding the offline systems’ bidirectional sequence stage on top of our features (unrestricted, or band-limited to their ± 10 -epoch effective receptive field) changes results by ≤ 1 point on all three corpora and ≈ 0 at matched training budget; (ii) the fully bidirectional twin gains only +0.9 offline; (iii) when *all* models must decide from the past only, the causal model is unchanged (online = offline by construction) while the bidirectional twin drops 3.7 points, losing to it by 5.0. The residual offline gap therefore lives in the epoch-level feature extractor—consistent with the stack results, where replacing spectral tokens by learned REVE features recovers +2.2 (P2018) and +1.4–2.0 BAC (HMC), on richer montages (≥ 4 channels, ≥ 200 Hz); on 2-channel 100-Hz Sleep-EDF, foundation features add nothing over 64-bin spectra.

TABLE III
WHEN DOES PRETRAINING HELP? PAIRED PRETRAINED-VS-RANDOM
COMPARISONS UNDER IDENTICAL FINE-TUNING. IN-DOMAIN PC:
PREDICTIVE CODING ON THE TARGET CORPUS. TRANSFER:
DONOR-CORPUS PRETRAINING, FULL WEIGHTS VIA PER-ELECTRODE
TOKENS.

Setting	Corpus (features)	Δ vs. random init
In-domain PC	SEDF-78 (tf64, 4 seeds)	+0.9 acc (n.s.)
In-domain PC	HMC (tf64, 8 seeds)	+0.0 $\pm$ 1.3 BAC
In-domain PC	P2018 (tf64, 994 subj.)	+0.001 κ
In-domain PC	HMC (REVE-B/L, 8 seeds)	-0.5 / -0.4 BAC (n.s.)
P2018 $\rightarrow$ SEDF transfer	SEDF-78 (tf64, 4 repl.)	+3.1$\pm$0.3 acc
P2018 $\rightarrow$ SEDF transfer	SEDF-78 (REVE-B, 3 seeds)	+0.3 acc (n.s.)

C. Pretraining helps only as an out-of-domain carrier

Table III condenses 20+ paired comparisons. In-domain self-supervised pretraining of the sequence decoder is null on every corpus and feature type—including at 994 training subjects, ruling out data scarcity as the explanation, and echoing in-domain nulls reported for BENDR and mulEEG [17], [18]. Two interventions do help, and by similar amounts: cross-corpus transfer of a per-electrode model pretrained on the largest corpus (+3.1 $\pm$ 0.3, every replicate positive, two independent donors—in the range reported for supervised transfer [3], [19]); and replacing spectral tokens with foundation-model features. Crucially they do not add: on REVE features the transfer effect collapses to +0.3, and the plain decoder on REVE features already sits where the spectral-token transfer arm lands. Both mechanisms inject structure learned *outside* the target corpus; once one supplies it, the other is redundant. This offers a unifying reading of the mixed SSL-for-sleep literature: pretraining pays in proportion to the out-of-domain knowledge it imports, not as in-domain regularization.

V. RELATION TO PRIOR WORK

Offline sequence models [1], [2], [3], [4], [5] own the leaderboards but read the future; RobustSleepNet and U-Sleep study cross-dataset robustness [20], [6]. EEG foundation models [7], [8], [9], [10], [11], [12] pretrain at scale but classify sleep epochs context-free (≤ 30 s inputs). Our result composes the two: frozen foundation features $\times$ causal sequence decoding, with the composition—not either part—setting the benchmark state of the art on HMC, in real time. Methodologically, all comparisons use released splits and protocols (NeuroLM’s HMC pipeline; SleepPyCo’s fold file), with pretraining corpora audited against target test subjects.

VI. CONCLUSION

A 3.5–11M-parameter causal decoder over per-epoch features is enough to (i) rival EEG foundation models from scratch, (ii) beat the strongest one on its own benchmark by adding the inter-epoch modeling it lacks—without fine-tuning it—and (iii) do so at one-epoch streaming latency. The measured price of causality (≈ 1 point) reframes real-time sleep staging as nearly free, and the pretraining analysis locates the value of self-supervision in out-of-domain structure. Scaling

the donor corpus (e.g., SHHS) and learned epoch encoders under the same causal recipe are natural next steps.

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